

SANGUINE

OS15 - Refining the System Development Approach Part 1: Group

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Sanguine is a physician-facing clinical interpretation layer for Mayo Clinic laboratory results. It aims **to** improve Mayo Clinic lab patient care outcomes at scale **by** augmenting physician-facing test results with relevant research **using** an agentic AI clinical librarian. In OS11 and OS13, we clarified the system boundary and identified the most important architectural decisions. In OS14, we quantified uncertainty and found two preferred architectures. Now, we choose a preferred concept after discussing with our sponsor and expand on it.

1: Preferred Concept Decomposition

1A: DSM

This DSM is also viewable at [this link](#).

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	Total
1 Identity and access manager	X	X	X	X	X		X	X												X	X	7
2 Privacy and access control layer	X	X	X	X	X		X	X														6
3 Lab results ingress	X	X	X	X																		3
4 Results normalization engine		X	X	X		X																3
5 Patient context repository	X	X				X																3
6 Case context builder				X	X	X	X	X	X	X	X	X	X	X	X							9
7 Internal evidence access API	X	X				X	X		X			X	X					X				7
8 External evidence access API	X	X				X		X	X			X	X					X				7
9 Evidence consolidation service						X	X	X	X	X								X				5
10 Evidence ranking module						X			X	X	X	X	X	X	X			X				7
11 AI role catalog										X	X	X										1
12 AI role orchestrator						X	X	X		X	X	X	X	X	X							7
13 Specialist interpreter						X	X	X		X	X	X	X	X	X							7
14 Cross-role reconciler										X	X	X	X	X	X							4
15 Interpretation composer						X				X		X	X	X	X	X		X		X		7
16 Review gateway															X	X	X	X	X			4
17 Release controller																X	X	X	X	X		4
18 Evidence trace store							X	X	X	X					X	X	X	X				8
19 Decision audit store															X	X	X	X	X	X		5
20 Physician delivery channel	X														X		X		X	X		5
21 Outcome feedback registry	X																	X	X	X		2
Total (# Related)	7	6	3	3	3	9	7	7	5	7	1	7	7	4	7	4	4	8	5	5	2	

Interfaces that needs monitoring Dependencies cluster region

Our current preferred concept is C1, "Maximalist". It both connects with our latest SPS review, to which we decided to dedicate the scope to the interpretation of clinical laboratory results for physicians, and emerges as one of the strong candidate concepts, remaining robust in the uncertainty situations presented in our prior work [1] [2]. This concept combines closed multimodal AI models that utilize the Gemini family, the research paradigm with RAG and model fine-tuning, a large pool of specialized roles, multiple internal and external clinical sources, and tiered release control [1] [2]. This means a physician-facing interpretation system that receives raw laboratory results, contextualizes them, grounds the source using multiple clinical references, coordinates with AI roles using multiple specializations for interpretation, and reviews the interpretation before delivery [1] [2].

To represent this preferred concept, we developed the L2 decomposition DSM of the product architecture, which brings together its main elements and their respective dependencies. At the L2 level, details are provided to expose the main interfaces, the change-propagation paths, and the coordination challenges of this concept. The L1 decomposition is shown in the appendix [3].

The L2 form decomposition reflects the flow of value creation in the preferred concept, similar to the function decomposition of OS14. The DSM shows our product architecture doesn't have a simple linear sequence. Instead, the DSM reveals 4 regions of dependency concentration, with the least concentrated at the top of the DSM and the most dense at the bottom: the interpretation region, the core function of the system, and the delivery. These strong concentration relationships show where changes affect other parts of the system most [2] [3].

The first region defines the system's structured entry and privacy boundary. The second region is where the system organizes the case and prepares the clinical references that will support the interpretation. The third and strongest

concentration indicates the heavy dependencies in multi-agent interpretation. The last region is dedicated to governance and delivery [3].

Our L2 decomposition is consistent with our OS14 conclusion, indicating that volatile AI agent functions should be modulated into more stable supporting components. These are elements of form, though they also align with some functions. Our uncertainty analysis reinforced the need to decouple AI functions instead of treating the system as a single monolithic mechanism. Therefore, our DSM reflects this conclusion by separating evidence management, function coordination, reasoning, synthesis, version control, auditing, and workflow delivery into distinct elements [2] [3]. The dependency relationships that deserve additional monitoring are: "Evidence consolidation service" with "Evidence ranking module" (interface between clusters), "AI role orchestrator" with "Specialist interpreter", "Cross-role reconciler" with "Interpretation composer", and "Interpretation composer" with "Release controller" (interface between clusters). These interfaces are in the most concentrated region of the DSM and have a direct impact on quality, multi-role consistency, release decisions, and the system's ability to deliver useful, reliable interpretation results. Also, changes to these interfaces are more likely to propagate to adjacent elements that affect evidence quality, interpretation consistency, release decisions, and physician delivery. The interface between clusters is more critical due to propagating beyond the local region, increasing the need for coordination [3] [6].

1B: Uncertainties in Elements and Dependencies

The most important uncertainties are concentrated in the denser bottom region of the DSM, where evidence handling, multi-agent interpretation, and release control interact. Our OS14 prior work already showed two main reasons why that region matters most: clinical accuracy was sensitive to AI role coordination, data sources, and release control. The second is that the cost per result is sensitive to release control because escalation creates queues and delays in human review [2] [5].

The first major uncertainty is the relationship among "Internal evidence access API", "External evidence access API," "Evidence consolidation service," and "Evidence ranking module" (#7-10). Our prior work in OS13 and OS14 showed the same tradeoff: more diverse sources can improve the clinical accuracy of the interpretation by enabling stronger grounding and comparison across different clinical sources, but they also increase harmonization problems such as duplication, conflicting sources, and retrieval inconsistency [1] [2] [4]. What remains uncertain is where the tradeoff stops being positive. In other words, we still don't know how much additional clinical reference diversity improves the final interpretation before the added filtering, harmonization, and conflict resolution limit the value gained. This is important because more comprehensive coverage only creates value if it improves the final interpretation enough to justify the extra integration efforts [1] [2] [4].

The second major uncertainty is the relationship among "AI role orchestrator," "Specialist interpreter," "Cross-role reconciler," "Interpretation composer," and "Release controller." Our prior work also showed that larger groups of AI roles can improve clinical accuracy at the cost of increased coordination variability, including conflicts across different interpretation specializations and duplicated results [1],[2]. Simultaneously, stronger review and release policies improve assurance and clinical accuracy but increase operational variation due to conditional human escalations and workload spikes. [1] [2] [5] [7] It is not yet clear what the proper balance is between the number of AI agents and the degree of human review required by the release policy, to preserve clinical accuracy and explainability without introducing escalations, delays, or excessive rework of the interpreted results that might compromise the system's viability. This uncertainty is particularly important because, if that balance is wrong, the system may still perform well in isolated cases but fail to remain scalable and cost-effective in practice. This risk is represented in threat #5 in our Q4 memo below. [6]

2:Hotspots (20%, one page)

The primary hotspots in our system are concentrated in two dense regions identified in the DSM: (1) the evidence management cluster and (2) the multi-agent interpretation and release control core. These regions are structurally central, exhibit high interface density (high C2), and have strong change propagation potential, making them the main drivers of system complexity and integration effort.

Hotspot 1: Evidence Integration and Ranking

The first hotspot lies in the interactions among internal and external evidence access, filtering and deduplication, and evidence ranking. This layer sits upstream of all reasoning functions and mediates between heterogeneous clinical data sources and downstream AI agents.

It exhibits high component complexity (C1) due to harmonization logic, retrieval strategies, and ranking algorithms. It also has high interface complexity (C2) because it connects to multiple data sources upstream and multiple reasoning modules downstream. Changes in source diversity, retrieval policies, or ranking criteria directly affect reasoning consistency and final outputs.

From a scope perspective, adding new clinical sources increases harmonization, filtering, validation, and testing effort across the cluster rather than simply adding a modular component. Because this layer conditions all downstream reasoning, instability here propagates broadly and increases integration burden.

Hotspot 2: Multi-Agent Interpretation and Release Core

The densest hotspot is the interpretation core, which includes AI role coordination, specialist reasoning, reconciliation, interpretation assembly, review gateway, and release control. This cluster represents the architectural center of the system and contains the highest concentration of interdependencies.

It combines high C1 (complex reasoning logic), high C2 (extensive coordination interfaces), and a tightly coupled topology. Changes such as adding specialist agents, modifying coordination protocols, or tightening release criteria can ripple across reconciliation logic, workflow sequencing, and review processes. Because elements depend on each other bidirectionally, small changes can create nonlinear increases in coordination effort and rework.

Since this cluster determines both clinical accuracy and scalability, misalignment here can significantly expand development scope.

Topological Implications

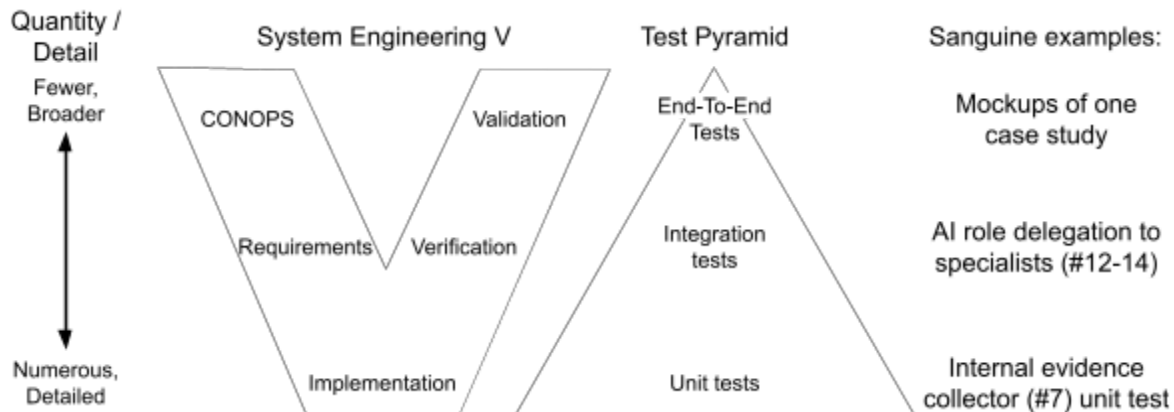
The overall architecture exhibits a core–periphery structure: relatively modular peripheral components surround a tightly connected interpretation core. Integration effort increases with interface density, strong clustering, and central positioning—all characteristics of the interpretation core. From a lecture perspective, the interpretation core behaves as a strongly connected component with high interface density, increasing architectural centrality and coordination load.

Such clustering amplifies nonlinear change propagation, where small local modifications can cascade across multiple dependent elements. As a result, testing complexity and iterative rework are expected to concentrate in this region, disproportionately increasing overall integration effort.

Therefore, the primary drivers of integration risk and scope expansion are the evidence management layer and the multi-agent interpretation core, where dependency concentration and change propagation potential are highest.

3: Validation and Verification (20%, one page)

Our V&V strategy connects the System Engineering V to the software engineering test pyramid. Both emphasize how systemic correctness traces back to earlier work. This figure shows both, with examples specific to Sanguine:



Any V&V strategy must start with the CONOPS and requirements. We drafted requirements back in OS9+10, but our understanding has evolved since then. At this stage, we'll validate with mockups of one case study. Case studies are common in medicine, so doctors would buy in, and mockups would be quick to build. These mockups will also be adapted into an end-to-end test later, showing traceability across the top row of the figure above.

The middle row is verifying the system meets requirements via integration tests. These will focus on the interfaces highlighted in 1A above, where we'll prototype the interpretation cluster. Finally, during implementation, each component will have unit tests, forming the bottom row of the test pyramid with component-level correctness.

3A: Sources of Technical Scope Growth

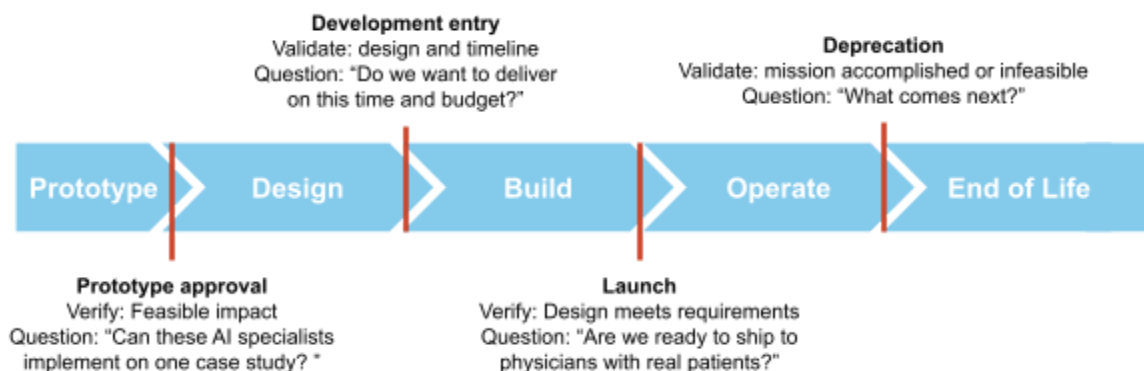
We foresee scope growth most in the interpretation cluster, especially AI role coordination (#12) through cross-role reconciliation (#14). These components have many red flags: they have the most connections, they have some of the interfaces highlighted in 1A, and they're close to their technical limits. This is why we're planning to build a prototype that includes only that cluster. Other scope growth areas could be inputs (#2-#5) if changes to privacy regulations or policies impose changes.

3B: Elements Near Limits

The LLM-based components in the interpretation cluster are near their technical limits in terms of tokens and context window. The frontier is advancing quickly! This creates uncertainty, as discussed in OS14 Q4, and motivates us to prototype them. The integrations with other health systems (release control #17, evidence access #7) could also be near technical limits because most EMR systems aren't incentivized to offer rich APIs, but we can address those with component-level work. Fortunately, none of the other elements are anywhere near technical limits.

3C: Lifecycle Milestones

This timeline shows PDP phases and stage gate reviews that separate them. Each milestone indicates what will be validated or verified, and shows how we'll increase confidence over time by answering a key question at each one.



MEMO

To: Dr. Chris Garcia, Mayo Clinic
 From: MITsdm Team 14a, Sanguine
 1 April, 2026

System Development Risk Assessment

Executive summary: We have identified risks to be aware of within both the developmental and operational phases of the project. In the developmental phase there are risks related to harmonization of data retrieval and significant risk of scope change related to interdependence, while the multi-agent interpretation can obscure failures and create upstream rework. In the operation phase of the lifecycle there are risks from market turbulence leading to shifts in model providers, regulatory shifts disrupting architectural and data governance decisions, and finally the user acceptance of the delivered outcomes.

Primary threats to technical scope and impact to lifecycle:

1. **Multi-Agent Interpretation Core (System Development Phase):** The bidirectional nature of the integration core and interconnectivity create risk of exception propagation that can introduce upstream rework or cascading failures. This bidirectional coupling within the interpretation core can create challenges in root causing the source of issues likely only found during later stages of verification. The decision to use a large group of agents also creates either constructive interference, where outcomes benefit from the diverse multi-agent interpretation, but poses a risk of destructive interference reducing the ability to discern with clarity.
2. **Data Harmonization (System Development Phase):** Our review of system architecture depicts risks due to interface complexity and topological complexity during electronic medical record and evidence retrieval. Risk grows as any error during the retrieval or interpretation process can propagate into the interpretation core.
3. **Market Turbulence (Operational Phase):** There are risks to pricing, availability, and model availability over the operation of the Sanguine. Being tightly coupled to a single model provider creates technical risk as it creates switching costs that must be borne if suddenly their model performance degrades, costs increase, or they are no longer available due to market consolidation or closure in operations.
4. **User Acceptance (Operational Phase):** There is the risk that the generation of clinical librarian data induces alert fatigue, or over availability of information reduces acceptance. This could shift during the operation and require system architectural rework to accommodate the physicians expectations.
5. **Regulatory (Operational Phase):** While monitoring of regulatory legislation can provide forewarning, large shifts to the regulatory environment create step function changes that could create sudden impactful risk to system architectural approaches. While best practices around decision transparency and data governance are to be deployed beyond current regulatory structures, new legislation could require a shift in architecture to regain compliance.

Further review and mitigation assessment:

- **Exception Injection:** Perturbation of our multiagent core (*bullet 1, left*) through exception injection during early development and prototyping phases can provide insights into the non-linear expectation progression that would be opaque. This is also beneficial in the data harmonization (*bullet 2, left*) stage to understand downstream dependency sensitivity and methods to reduce that dependence.
- **Targeting Prototyping:** Prototyping of key standalone components during the development phase allows for better understanding of the standalone component without interface complexity, useful in understanding whether data harmonization (*bullet 2, left*) is likely to introduce errors to downstream interpretation core (*bullet 1, left*).
- **Modularity of AI integration:** Modularity within the agentic roles allow for optionality in selecting a provider to fulfill that role. Market turbulence (*bullet 3, left*) can generate sudden positive shifts to capitalize on where model performance improves, inversely lagging performers or sudden market departures could expose dependence risk that modularity could resolve. This also decreases switching costs allowing for lower sensitivity to pricing from model providers.
- **A/B Testing and User surveys:** Development of a framework for A/B testing tied to user feedback surveys allows for directional guidance to mitigate issues of user acceptance (*bullet 4, left*). This can be continued into the operational stage of the product lifecycle to ensure changes in preference are met and quality is maintained.

Summary: The current risks posed during different stages of the product lifecycle can create technical rework and rearchitecture if not mitigated using the strategies proposed. Different tactics at different parts of the lifecycle can be employed to keenly address the risks posed and ensure stability to unforeseen or minor risks not addressed in this memo.

Appendix 1

Decomposition L1, adapted from L1/L2 decomposition from OS13.

1. Access and Trust Boundary

This group includes the parts of the architecture that control system access and permissions and how consent and privacy are applied.

2. Results Reception and Case Framing

This group includes the elements that receive the laboratory results, perform the normalization and integration into the patient's context to prepare the case for interpretation.

3. Evidence Base

This group includes the parts that connect to internal and external clinical reference sources and organize the evidence that will support the interpretation.

4. Multi-Agent Interpretation Core

This group includes the elements where specialized AI roles are defined, coordinated, reconciled, and results are assembled to produce the final clinical interpretation.

5. Control, Review, and Release

This group includes the elements that manage review and release decisions, as well as the traceability needed to provide audit trails of the interpretation process.

6. Clinical Delivery and Learning

This group includes the elements that deliver the interpretation into the clinical workflow and capture user feedback for later refinement and learning.

Appendix 2

References

- [1] Team 14A, OS13 – System Architecture, MIT SDM EM.413, Feb. 2026.
- [2] Team 14A, OS14 – Identifying Threats to Technical Value Delivery, MIT SDM EM.413, Mar. 2026.
- [3] Team 14A, OS11 – System Architecture, MIT SDM EM.413, Feb. 2026.
- [4] F. Neha et al., “Retrieval-Augmented Generation (RAG) in Healthcare,” AI, vol. 6, no. 9, 2025.
- [5] O. O. Olakotan and M. M. Yusof, “Evaluating the alert appropriateness of clinical decision support systems in supporting clinical workflow,” Journal of Biomedical Informatics, vol. 106, p. 103453, 2020.
- [6] Mayo Clinic Laboratories, “Commercial laboratories,” 2026.
- [7] Z. Ge, H. Li, Y. Wang, N. Hu, C. J. Zhang, and Q. Li, “ClinicalAgents: Multi-Agent Orchestration for Clinical Decision Making with Dual-Memory,” arXiv:2603.26182, 2026.

AI Disclosure

Google’s Gemini was used to review drafts of this assignment and act as the role of a MITsdm TA providing feedback on sections with strong arguments and areas requiring improvement. All work in response to this feedback is the author's own.